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2.10 Data and MI Design

Where every field is captured in the target state, how protected fields are preserved, how history is mapped, and what the platform must report

Deliverable	2.10 Data and MI design: to-be capture points, new data elements, protected-field preservation, deal-type mapping, migration rules, KPI definitions on the to-be stage model, MI products
Version	0.6 – complete against 2.3 / 2.4 / 2.8 (Week 11). Field-level lineage rows depend on 1.8 v1.0 (DR-10); v1.0 at Design Authority (Week 12) with Head of MI & Regulatory Reporting acceptance
Derived from	1.8 lineage and protected fields; 1.4 stage model and metric dictionary; 2.3 maps; 2.4 rule inputs; 2.5 regulatory statement; 2.6 GD-01 to GD-08; 2.8 DAT-01 to DAT-08, INT-08, NFR-04
Owner	WS3 Data & MI (Data / MI Analyst) with Bank BA
Accepted by	Head of MI & Regulatory Reporting; named field-group owners; Replacement Programme Data Lead (DD-09, migration rules); second line (regulatory preservation)

1. Design principle applied and what this document guarantees

DP-09: capture once, at source, with a named owner; protected fields are a constraint. This document is the evidence that the target design honours it. It makes four guarantees, each verifiable by a table below:

#	Guarantee	Verified by
1	Every protected field group (1.8 §6) has a to-be capture point at the same or an earlier step, with the same definition, and an owner	§2 capture map; §3 preservation table
2	Every data element a rule in 2.4 reads is captured before the rule evaluates it, in structured form	§4 new elements, keyed to rule inputs
3	History remains reportable across the deal-type change and the ESR Cleanse exclusion, and baseline metrics remain comparable across the stage-model change	§5 mapping and migration rules; §6 stage mapping
4	Every KPI, control measure and benefit in 2.11 / 2.13 can be produced from platform data without a survey	§6 metric definitions; §7 MI products

Ownership model (from 2.9 §5)

Field group	Accountable owner	Definition change requires
F10 forbearance; F11 ERV, provisions, write-offs, categorisation	Head of SBS (with Head of MI for reporting definition)	Owner + MI + second line
F06 risk grade and classifications	Head of Credit Risk Policy	Owner + MI
F04 facility terms; F05 exposure; F08 security	Head of Lending Operations (F08 with Lending Services)	Owner + MI
F01 customer; F02 parties	Head of Lending Operations (KYC content: Compliance via CIF)	Owner + Compliance
F07 decisions and authority; F03 deal and pattern; F12 ownership; F13 documents; F14 workflow events; F15 deposits	Head of Lending Operations	Owner + MI
F09 covenants and conditions	Head of Credit Risk Policy	Owner
Lineage register and protected-fields list	Head of MI & Regulatory Reporting	Change control; annual attestation

2. To-be capture map

Every field group from 1.8 §5 with its as-is and to-be capture point, mode, and what changes. Protection level from 1.8 §6 (P1 regulatory · P2 regulatory-derived / audit · P3 MI continuity · P4 operational).

#	Field group	Prot.	As-is capture (1.8)	To-be capture (2.3 step)	Mode to-be	Captured by	What changes	Rules reading it (2.4)
F01	Customer identity, counterparty type, sector, SME classification	P2	NM-1 S3–S5 manual; CIF for existing	SC-1b S3 (CIF match / create); SC-1a S2 (loaded)	Pull-through	CIF; platform	Captured once in CIF; never re-keyed; sector and SME classification pulled, not typed	RS-01 c.10, RS-05, RS-14
F02	Parties, guarantors, connection membership	P2	NM-1 D1 loop; VA-1 S5; CS-2	SC-1a S4 (completeness rule); SC-1b S3; P3 V-07	Pull-through + rule	Platform; RM	Completeness enforced; connection traversal for RS-11 i.7	RS-11, RS-03 V-07
F03	Pattern, deal type parameter, route, dates, restructure flag	P3	NM-1 S7; RN-1 S5; SR-1 S6; SD-1 S6	SC-1a S5 / SC-1b S4 (pattern and parameters); SC-2 D1 (route)	System-set	Platform	**New:** route, eligibility result, failing conditions; deal type becomes a parameter; mapping table (§5)	RS-01 c.9, c.12; RS-10
F04	Product, limit, rate and margin, fees, repayment, expiry, tranche schedule	P1	NM-1 S8; SL-1 S8–S9; SR-1 S7–S8; SD-1; CS-2 S1	SC-1b R2 (P1); SC-1a R1 (P2 structural validation; P3 change capture); SC-2 S1 (P4 allocation); SC-2 S5 (approved terms)	Manual + system	RM; platform	Approved terms become the document source (no divergence between system and letter); tranche schedule structured; limit arithmetic enforced; balances via INT-01	RS-01 c.1–2, c.11; RS-02; RS-05; RS-06
F05	Exposure: total, personal, NAG, group	P1	Derived; ESR Cleanse updates	Derived from F04 across connection at SC-2 S3; migration rule for personal / NAG (§5.3)	Derived	Platform	ESR Cleanse removed as a capture path; migration carries the updates; derivation documented	RS-05; RS-01 c.11
F06	Risk grade, grade date, ECRS, lending category, debt-servicing result	P1	*-2 R3; off-platform W-10	SC-2 R2 (R-FULL) / R1 (R-FAST refresh per RS-04); grade history retained	Manual (on-platform) + integration (R2)	RM; Credit	Grade date and history structured (RS-01 c.4–5 depend on it); analytics evidenced on platform; staleness rule	RS-01 c.3–5; RS-04; RS-05; RS-11 i.2; RS-08 re-validation
F07	Decision, authority level, approver, rule version, decline reason	P2	*-2 D2, D4, S5; EX-01	SC-2 S3 (authority), R3 / D3 (decision), S5; EX-A S1 (reason)	System-written	Platform	**New:** authority level, matrix version, rule evidence, decline and withdraw reason codes, sampling tag and finding	RS-05 (writes); CE-001 sampling
F08	Security type, value, valuation date, allocation, perfection status, registration, release	P1	EX-05; RF-2; NM-3 S5	SC-3 S4 (instruction from conditions), G1–G2 (status, evidence); TB-P3R (release sequence)	Manual + system	Lending Services; platform	Registration evidence attached; release sequence enforced; shared-security relationship explicit	RS-04 (valuation age); RS-07; RS-02 t.8
F09	Covenants, tracking items, waivers, conditions precedent	P4	VA-3 S4; RF-2 S4; batch W-03	SC-2 S5 (conditions structured); SC-3 S5 (CP items); native lifecycle (OP-24, R2)	System + manual	Credit; RM; platform	Conditions structured at decision; CP items generated; waiver lifecycle native with audit	RS-01 c.6; RS-11 i.3; RS-08

F10	Forbearance status, date, measure	P1	CR-2 S7; batch W-02	P5 B2 (assessment); native status logic	Manual + system	SBS; platform	**Definition, capture step and timing unchanged.** Batch logic reproduced natively; cut-over parallel run (§5.4)	RS-11 i.4
F11	ERV, provisions, write-offs, categorisation cause, recovery team	P1	CS-2 S4; CR-2 S5–S6	P5 S2 (categorisation), B3 (structured memo adds), B4 (recovery fields with validation)	Manual (validated)	SBS	**Definition unchanged.** Validation rules on entry; memo adds structured fields, replaces none	— (reporting only)
F12	Officer, segment, team, ownership history	P2	Deal creation; CR-2 S8; batch W-04	SC-1 (from user / event); P5 S2 (transfer record)	System-written	Platform	**New:** ownership history with dates and reason; segment derivable at any point in time (MA-03 retired)	RS-13
F13	Document type, status, dates, signature evidence, reconciliation result	P4	*-3 S1–S3 (partial; DR-08)	SC-3 S1–S3; EX-E (chase, extension, re-validation)	System-written	Platform	**New, captured for the first time:** issued / viewed / signed / returned timestamps; chase log; extension reason; re-validation record	RS-06; RS-08
F14	Workflow stage, timestamps, actor, task open / close, queue	P3	Every transition (partial coverage)	Every transition and task (DAT-06); stage model §6	System-written	Platform	Complete coverage; task time; mapped to baseline stages	SLA timers; all T-metrics
F15	Deposit opportunity status, value, reason, pricing exception	P4	NM-1 S6; DP-1	Per SD-03: SC-1 status or CRM; reason code; exception approval	Manual	RM	**New:** reason codes; pricing-exception record	—

3. Protected-field preservation statement

For each P1 and P2 field group: the design obligation from 1.8 §6.1, how the design meets it, and the cut-over test. This section is the evidence for 2.5 §4.1 and DD-09.

Group	Lever(s) touching it	Obligation and how met	Cut-over test	Sign-off
F10	L7 retire batch; L10 distress	Status definition unchanged; batch rules re-implemented as native logic from the job specification (DR-09); assessment captured at P5 B2 as today	One reporting cycle parallel: batch status vs native status per connection; zero unexplained differences	Head of SBS; Head of MI; second line
F11	L14 capture once; L15 structured memo	Fields unchanged; validation added (ranges, mandatory on categorisation); memo adds fields, replaces none; narrative retained	Field-level reconciliation of migrated values; validation rules tested on historic cases	Head of SBS; Head of MI
F04, F05	L1 fast-track; L5 triage; L6 ESR Cleanse; L12 patterns	Terms captured once and used as document source; exposure derived by documented rule; ESR Cleanse updates specified for migration (§5.3); fast-track re-confirms limit and dates (RS-01 c.1–2)	Exposure recalculated post-migration vs pre-migration per connection; variance explained	Head of Lending Ops; Head of MI
F06	L1 fast-track; L11 submission; L14	Grade and classifications captured at SC-2 on every route; R-FAST requires grade ≤ 15 months (RS-01 c.4) and refresh per RS-04; history retained for RS-01 c.5 and RS-11	Grade history migrated; staleness rule tested on cohort	Head of Credit Risk Policy
F08	L8 security instruction; L15	Values and dates captured at instruction and completion; registration evidence attached; release sequence enforced	Security records reconciled; shared-security links verified	Head of Lending Services
F01, F02	L2 deal start	Captured in CIF; pulled through; completeness rule adds, removes nothing	CIF reference present on 100% of migrated customers	Compliance; Head of Lending Ops
F07, F12	L3 automated DA; L14	Authority level and rule version recorded on every decision (today: approver name only); ownership history retained	Migrated decisions carry approver; new decisions carry level; attempted-breach log operational	Second line; Internal Audit

Statement for second line: no P1 field changes definition, capture step or timing. Four groups gain additional capture (validation, history, evidence) and none loses any.

4. New data elements

Elements the as-is does not hold in structured form and the target design requires. Each is keyed to the rule or control that reads it, so none is captured without a consumer (DP-09).

ID	Element	Type / values	Captured at	By	Consumed by	Group
N-01	Variation reason code(s)	Code list V-01..V-12; multi	SC-1a R1	RM	RS-03, RS-06, RS-07; CE-003 exception report	F03
N-02	Route selected; eligibility result; failing conditions; rule version	Enum + structured	SC-2 D1	Platform	SC2-04 evidence; sampling; cohort MI; T-metrics by route	F03 / F07
N-03	Required authority level; approver level; DA matrix version; override reason	Enum; text	SC-2 S3, R3	Platform; approver	CE-009; control MI (attempted breaches)	F07
N-04	Decline reason code; withdraw reason code; appeal outcome	Code lists	EX-A S1; SC-2 D1; EX-A D1	RM; Credit	RR-07; R7; CE-006	F07
N-05	Deferral request: items, owner, due date, loop counter	Structured	EX-B S1	Credit; platform	RS-09; R1, R2, R3; RS-04 maintenance	F14
N-06	Sampling selection flag; finding category; reviewer	Flag; enum; user	SC-2 K2	Platform; Credit	CE-001/003 MI; suspension trigger	F07
N-07	Early-warning flag result and indicator values	Boolean + indicator set	SC-1a S3; SC-2 submission	Platform	RS-01/02/03 exclusion; P5 trigger; CE-005	F06 / F10
N-08	Document status and timestamps; signature certificate; reconciliation result	Enum + timestamps	SC-3 S1–S3	Platform; e-sign service	RS-08; T5; R4; CE-004	F13
N-09	Chase log; extension reason; 3-month re-validation record	Structured	EX-E	Platform; RM	CE-002; MI letters outstanding	F13
N-10	Conditions precedent items with evidence; gate status	Structured + attachments	SC-2 S5 (conditions); SC-3 S5, R1	Credit; RM	C9; OP-13; S4 timing	F09
N-11	Tranche schedule (date, amount, conditions, status)	Child records	SC-1b R2 / P1	RM	P4; RS-02 t.7; C19	F04
N-12	Referral record: trigger, summary, linked deals and their status, authority decisions	Record type	P5 S1, K1, B1	RM; Credit; SBS	T8; C13, C14; cohort MI	F11
N-13	Ownership transfer: from, to, date, reason, approver	Record	P5 S2; admin	Platform	C26, C27; segment attribution	F12
N-14	Post-approval change: fields changed, materiality class	Structured	EX-C R1, S1	RM; platform	RS-12; R5	F04 / F07
N-15	Task open / close timestamps; queue; SLA status	Timestamps	Every task	Platform	W1, W2, W3; SLA MI; benefits tracking	F14
N-16	Pre-screen outcome and code	Enum	SC-1b S2	Platform	RS-14; C0; decline MI	F07
N-17	Lead record: source, created, first contact, converted	Record	SC-1b S1	RM / intake	S0 timing; conversion MI	F03
N-18	Deposit reason code; pricing-exception approval	Enum; record	DP per SD-03	RM	MI-06	F15

Eighteen new elements. None is free text where a code is reportable; each has a named consumer. Elements N-02, N-03, N-06, N-08 and N-09 are the evidence base for the equivalence register; without them 2.5 cannot be operated.

5. History, mapping and migration rules

5.1 Deal-type mapping table (DAT-03)

As-is deal type	To-be pattern	deal_type parameter	Additional derivation	History rule
New	P1	New	—	Direct
Increase	P1	Increase	—	Direct
Renewal	P2	Renewal	route = Full (historic)	Direct; historic route null → Full for cohort comparison
Extension Pending Review	P2	EPR	counter from sequence (DR-04)	Direct; counter back-filled where sequence identifiable
Variation	P3	Variation	reason = Unclassified unless derivable	Where DR-04 fields-changed data exists, derive V-01..V-09; removals (zero limit, all facilities) → V-11; else Unclassified
Remove All Facilities (un-typed)	P3	Variation, reason V-11	—	Re-coded from Variation by the removal signature
CAP Referral	P5	Referral	—	Direct
CAP Strategy	P5	Strategy	—	Direct; linked to referral where identifiable
Shadow Limit Reallocation	P4	Reallocation	facility_type = Shadow	Direct
Staged Drawdown + request type Variation	P4	Allocation	facility_type = Staged	Re-coded from the field combination (W-16)
Staged Drawdown (origination; 0 deals)	—	—	—	Type retired; no records expected
ESR Cleanse	— (excluded)	—	flag = migration_record	Excluded from all volume and cycle MI; retained as records for audit; exposure effects carried by §5.3
DO NOT USE	— (excluded)	—	flag = invalid_type	Any records re-coded by inspection or flagged

Retention: mapping table maintained for 24 months after cut-over (NFR-08); all MI keyed on pattern + parameter from go-live, with a “legacy type” column available for reconciliation.

5.2 Migration population rules

Population	Rule	Reconciliation
Closed deals (history)	Migrated with mapping table applied; stage timestamps mapped per §6.2; no re-processing	Counts by legacy type = counts by pattern + parameter
Open deals in SC-1 / SC-2 equivalent	Migrated into the to-be stage from the as-is status (§6.2); route = Full; rules evaluated on next action (no retrospective fast-track)	Open count and stage distribution before / after
Open deals awaiting letters	Migrated into SC-3 with document status = issued (date from DR-08 or RM confirmation); EX-E timers start from original issue date	Letters outstanding count
Open referrals / strategies	Migrated as referral records with linked deals; F10, F11 values carried field for field	§5.4 parallel run
Facilities, security, covenants	Migrated as entities with relationships; shared security links derived from allocation data	Entity counts; exposure recalculation

5.3 Exposure update rule (SD-01 option c; DAT-04)

ESR Cleanse records update personal exposure on personal-associated deals and external NAG exposure on business records. The migration specification (WS3 to Programme Data Lead, W7) states, for each exposure field: source record, derivation, timing, and the equivalent derivation in the to-be data model so that the update happens by rule rather than by a deal. Post-migration exposure per connection is recalculated and compared with the pre-migration value; variances above a tolerance set by Credit Risk are investigated before go-live.

5.4 Regulatory parallel run (DAT-05; CE-015)

Element	Parallel run	Pass criterion	Owner
Forbearance status (F10)	Batch job W-02 and native logic both run for one monthly cycle on the migrated population	Zero unexplained status differences; explained differences documented and accepted by SBS and MI	Head of SBS; Head of MI
Waiver expiry (F09)	Batch W-03 vs native lifecycle, one cycle	As above	Head of Credit Risk Policy
Regulatory extract (INT-08)	Extract from as-is and to-be for the same period	Field-level match on all P1 groups; differences explained by mapping only	Head of MI & Reg Reporting

6. KPI definitions on the to-be stage model

6.1 To-be stage model and mapping to baseline (GD-08)

Baseline stage (1.4 §2)	To-be stage	Start event (to-be)	End event (to-be)	Comparability
S0 Pre-deal	S0	Event / lead created	Deal opened	New: measurable for all event-triggered deals (baseline had partial coverage)
S1 Preparation	S1	Deal opened	Submitted (standard met; DA determined)	Direct
S2 Decision	S2	Submitted	Decision recorded	Direct; add split by route
S2a Credit queue	S2a	Credit task created	Task opened by assessor	Direct (task timestamps now complete)
S2b Credit assessment	S2b	Task opened	Decision recorded	Direct
S3 Documentation	G-DOC	Decision approved	G-DOC met	Direct
S3a Letter out	G-DOC a	Letter generated	Issued	Direct (now same-day)
S3b Customer	G-DOC b	Issued	Returned	Direct; now complete coverage
S4 Completion	Gates	Decision approved	All applicable gates met	Direct; add G-SEC and G-CP separately (new)
S5 Fulfilment	S5	Fulfilment task created	Fulfilled	Direct
S6 SBS	P5	Referral record created	Categorised / returned	Direct

6.2 Status-to-stage mapping for migration

The as-is status codes (1.4 Appendix A, agreed with MI in Week 3) map to to-be stages as above; open deals are placed by this mapping at cut-over. The mapping is held in the lineage register and re-used by MI for the 24-month comparison window.

6.3 Metric definitions carried forward and added

ID	Metric (1.4)	To-be source and definition	New cuts	Used by
V1–V5	Volumes; Credit-routed share; fast-track eligible	Deals by pattern + parameter; route field; eligibility result (N-02)	Route; segment	2.11; cohort MI
T1–T8	Cycle times	Stage timestamps per §6.1; calendar days; P50 / P80 / P95	Route; pattern; segment	2.11 targets; dashboards
W1–W3	Touch, wait, flow efficiency	Task open / close (N-15) replaces survey for touch; wait = stage elapsed − task time	Role; stage	Benefits tracking without survey
R1–R8	Rework and quality	Deferral counter (N-05); reset reason (N-09); R5 from N-14; R8 from timer resets; decline / withdraw from N-04	Route	2.11; CE-001/003 MI
K1–K3	Cost	Task time × DR-16 rates	Pattern; role	2.13; benefits
Added for the target state				
C1	Attempted approvals below authority	Count from N-03 log	Segment; role	O3 control MI

C2	Sampling: completion rate; finding rate by category	N-06	Segment; route; reason code	CE-001/003; suspension trigger
C3	Cohort outcomes: 12-month downgrade, arrears, referral rate, fast-track vs full	N-02 joined to F06 history, conduct, N-12	Grade band; segment	Quarterly to Credit Risk and second line
C4	Administrative variations that changed a monitored field	N-01 vs F04 / F08 / F02 deltas	Code	CE-003 exception report
D1	Letters outstanding > 8 weeks; extensions; re-validations; resets	N-08, N-09	Pattern	CE-002 MI
E1	Early-warning flag rate; flagged-to-referral conversion; T8	N-07, N-12	Indicator	RS-11 threshold review
S1	SLA attainment per stage and owner	N-15 vs SLA table	Team	2.9 §4.1

7. MI products

ID	Product	Audience	Cadence	Content	Source
MI-P1	Pattern dashboards (five)	Business Owner; heads of segment	Weekly	1.4 §6 template: volumes, T1 / T2, stage bars touch vs wait, rework, segment cut, hypothesis / target read-out	Platform extract (INT-08)
MI-P2	Queue and SLA board	Team leaders; Credit manager; LS; Fulfilment	Live / daily	Queue depth, ageing, SLA status by team; escalations	Platform native
MI-P3	Control MI	Head of Credit; second line; Audit	Monthly	C1 attempted breaches; C2 sampling; C4 exception report; D1 letters; rule-version changes	Platform extract
MI-P4	Cohort outcome report	Credit Risk Policy; second line; Risk Committee	Quarterly	C3 fast vs full outcomes by band and segment; E1 flag performance; threshold recommendations	Platform + conduct + grade history
MI-P5	Benefits tracker	SteerCo; Finance; benefit owners	Monthly (Phase 4)	KPI vs baseline vs target per 2.11; capacity by role from task time; realisation status	MI-P1 inputs; DR-16
MI-P6	Regulatory extract	Reg Reporting	Per return	All P1 / P2 fields per lineage register; unchanged definitions	INT-08
MI-P7	Decline and conduct MI	Lending Ops; Conduct	Monthly	N-04, N-16 by stage, code, segment; appeals	Platform extract
MI-P8	Rule-maintenance pack	Credit Risk Policy; Lending Ops	Quarterly	Per rule set: evaluation volumes, outcome distribution, override rate, sampling findings attributable, proposed changes	Rule evidence log (RUL-02)

7.1 Manual adjustments retired

Adjustment (1.8 §4)	Cause	Retired by	When
MA-01 Re-classify removals recorded as Variation	No removal type	N-01 reason V-11; mapping table for history	Go-live
MA-02 Exclude ESR Cleanse from volume MI	Migration recorded as deals	§5.1 exclusion; no new records	Wave 0 (now)
MA-03 Segment attribution after officer-code batch	No ownership history	N-13 ownership history	Go-live
MA-04+ (from DR-10 interview)	[to be listed at 1.8 v1.0]	[mapped to N-elements or lineage fixes]	

8. Dependencies, risks and sign-off

Dependency	Needed for	Status
1.8 v1.0 field-level lineage (DR-10)	§3 cut-over tests at field level; MA-04+	Pending W4–6
Batch job specifications (DR-09)	§5.4 native logic reproduction	Pending W4
Platform data model (CAP-06, 2.7)	Confirmation that N-01 to N-18 are configurable	FG-1 / FG-3
Programme migration tooling	§5.2, §5.3	Programme
1.4 Appendix A status mapping	§6.2	Agreed W3

Risk	Mitigation
Variation history largely Unclassified, weakening the P3 baseline comparison	50-deal sample classification (1.4 step 7) gives the historic mix; cohort comparison for P3 uses post-go-live data
Conduct data unavailable for C3 cohort outcomes in Release 1	Downgrade and referral rates used; arrears added when INT-01 lands
Parallel run reveals batch logic undocumented behaviours	DR-09 specification reviewed with RPA / BLP support before build; one extra cycle budgeted

Sign-off

Role	Scope	Name / date
Head of MI & Regulatory Reporting	Whole document; lineage and protected-fields ownership	
Head of SBS	F10, F11 preservation; §5.4	
Head of Credit Risk Policy	F06, F09 preservation; C3 definitions	
Head of Lending Operations	F03, F04, F07, F12–F15; N-elements	
Replacement Programme Data Lead	§5 mapping and migration rules (DD-09)	
Senior Operational Risk Manager	§3 statement	